

K2-29b

R-2224

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-29_211225 · created 2026-08-15T00:16:22+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459573.726 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.049
Transit depth (Rp/Rs)^2	2.86 +/- 1.66 [%]
Orbital Inclination (inc)	88.65 +/- 2.38
Airmass coefficient 1 (a1)	1.12 +/- 0.59
Airmass coefficient 2 (a2)	-0.09 +/- 0.52
Scatter in the residuals of the lightcurve fit is	51.78 %
Best Comparison Star	None
Optimal Aperture	1.8
Optimal Annulus	4.94
Transit Duration (day)	0.044 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.169 ± 0.049 vs 0.1419 (deviation 0.55σ)
Duration fit vs published	0.044 d vs 0.0925 d (deviation 4.41σ)
Mid-time O–C	-11.2 min
Depth SNR	3.4
Residual scatter	51.78 %

Light curve

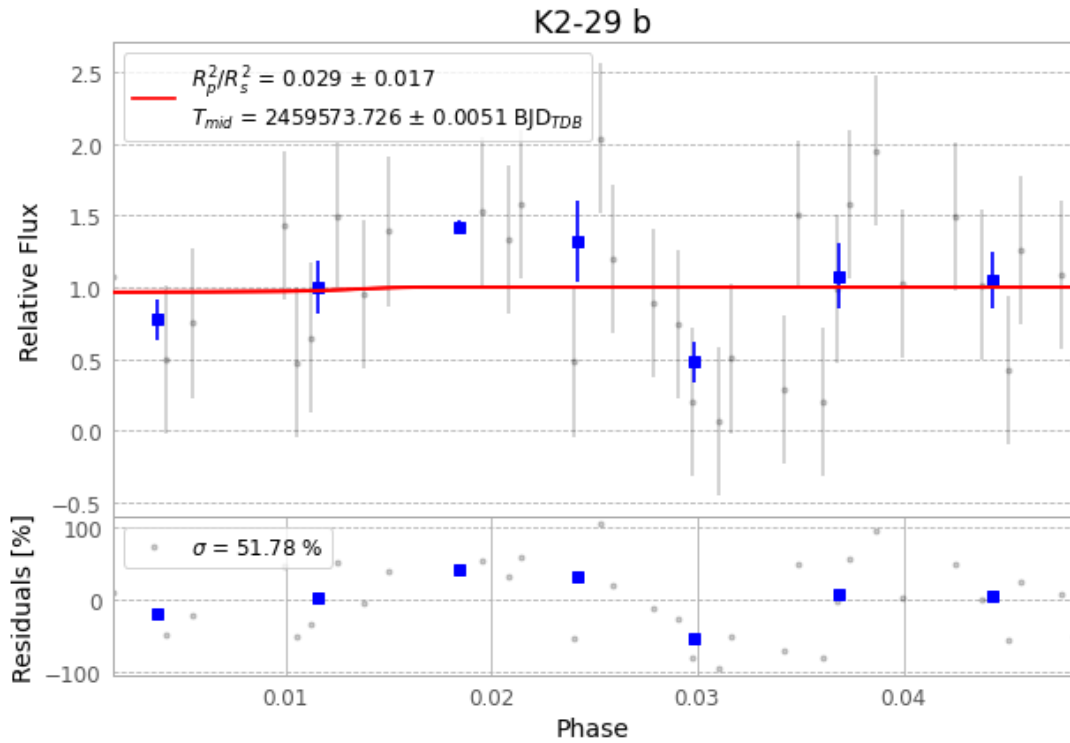


Figure 1 — FinalLightCurve_K2-29 b_2021-12-24.png (SHA-256 bd81689bb744947d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [566, 438], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d839bb72cbfd95ec...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

78 FITS frames (51 MB), K2-29211225052112.FITS → K2-29211225091212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-K2-29-211225.log (91678e44d2fc...), FinalLightCurve_K2-29 b_2021-12-24.png (bd81689bb744...), FinalLightCurve_K2-29 b_2021-12-24.csv (e84d903bf63e...)

Compute resources

Wall time 128.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-15 00:16Z.